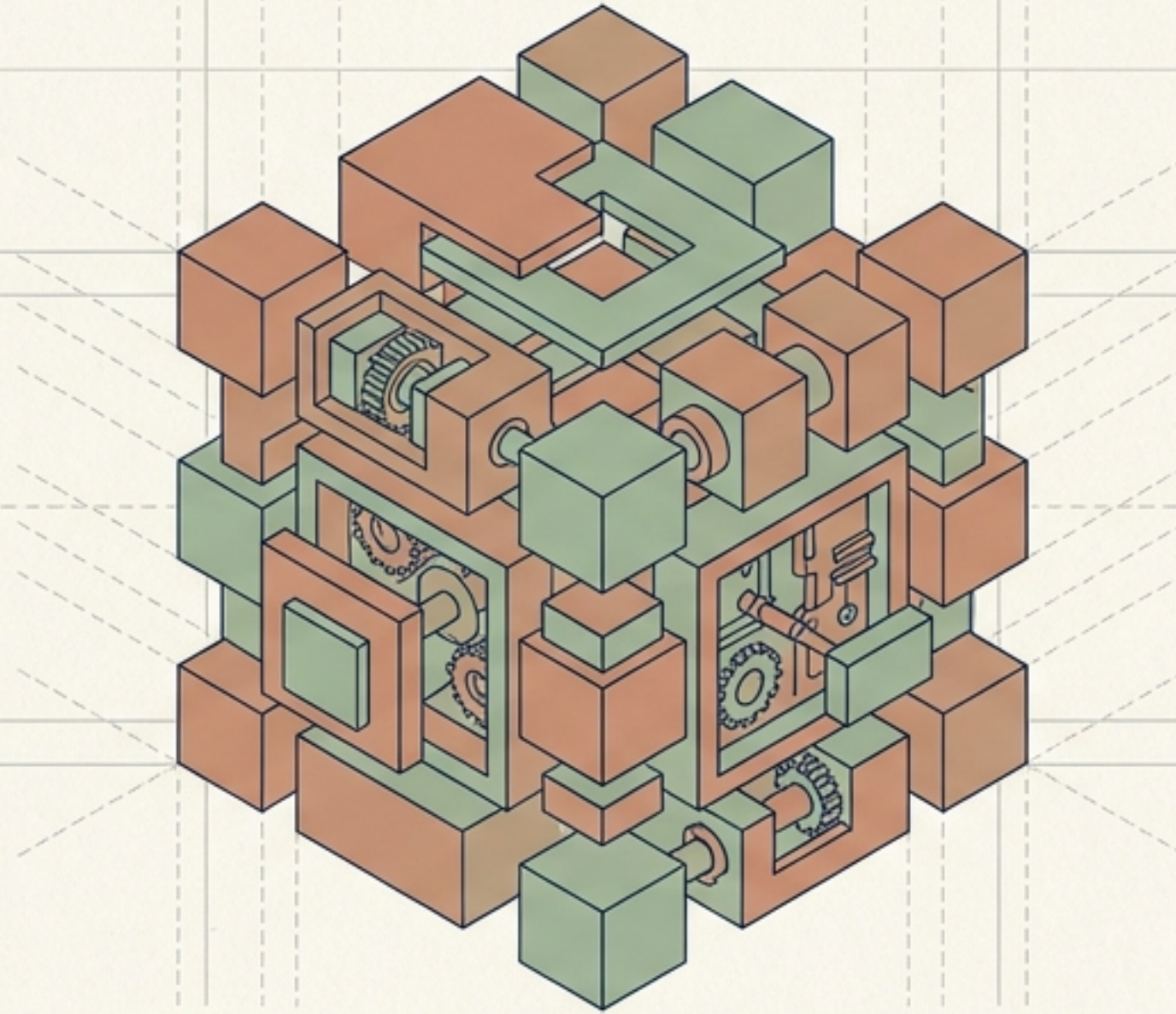
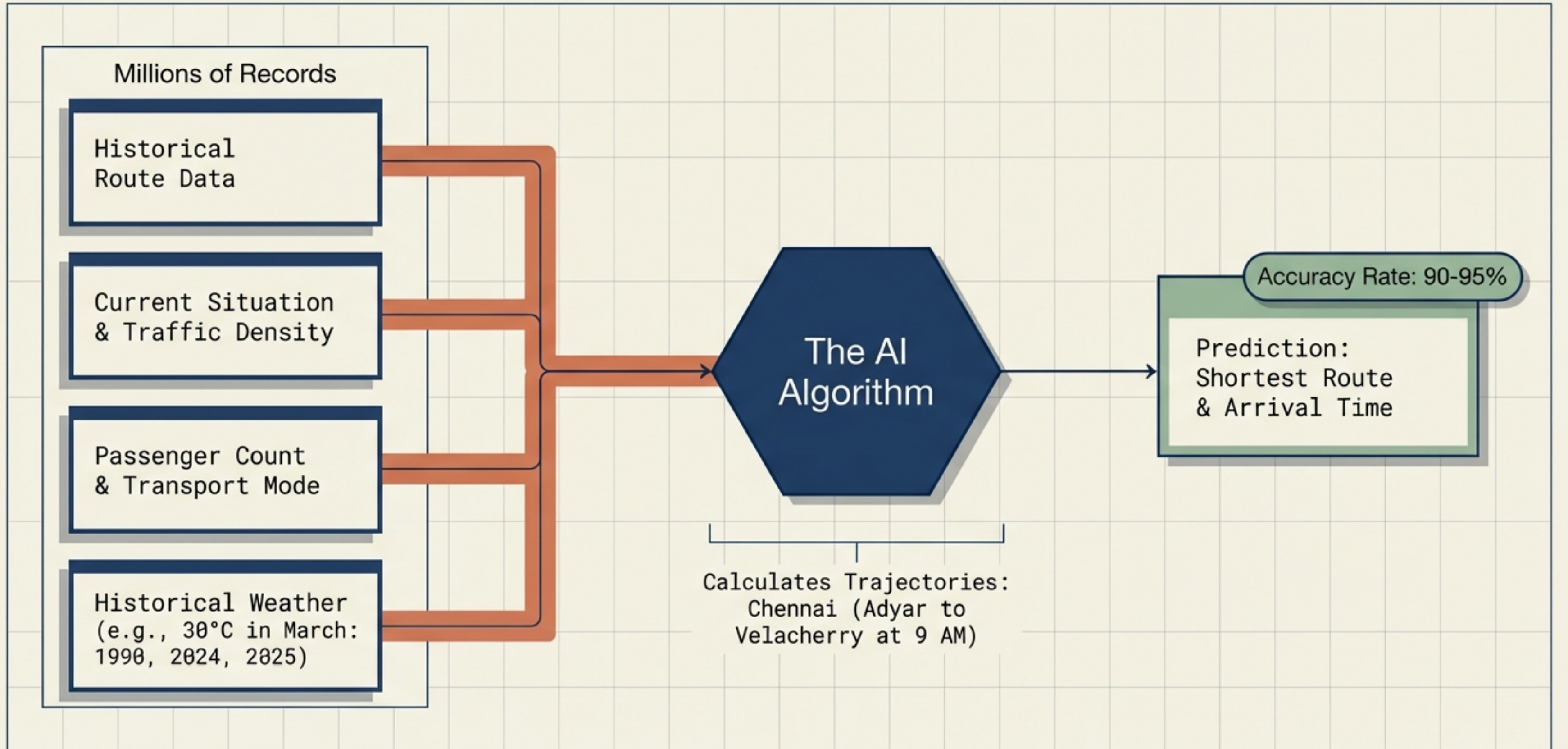


The Architecture of Intelligence

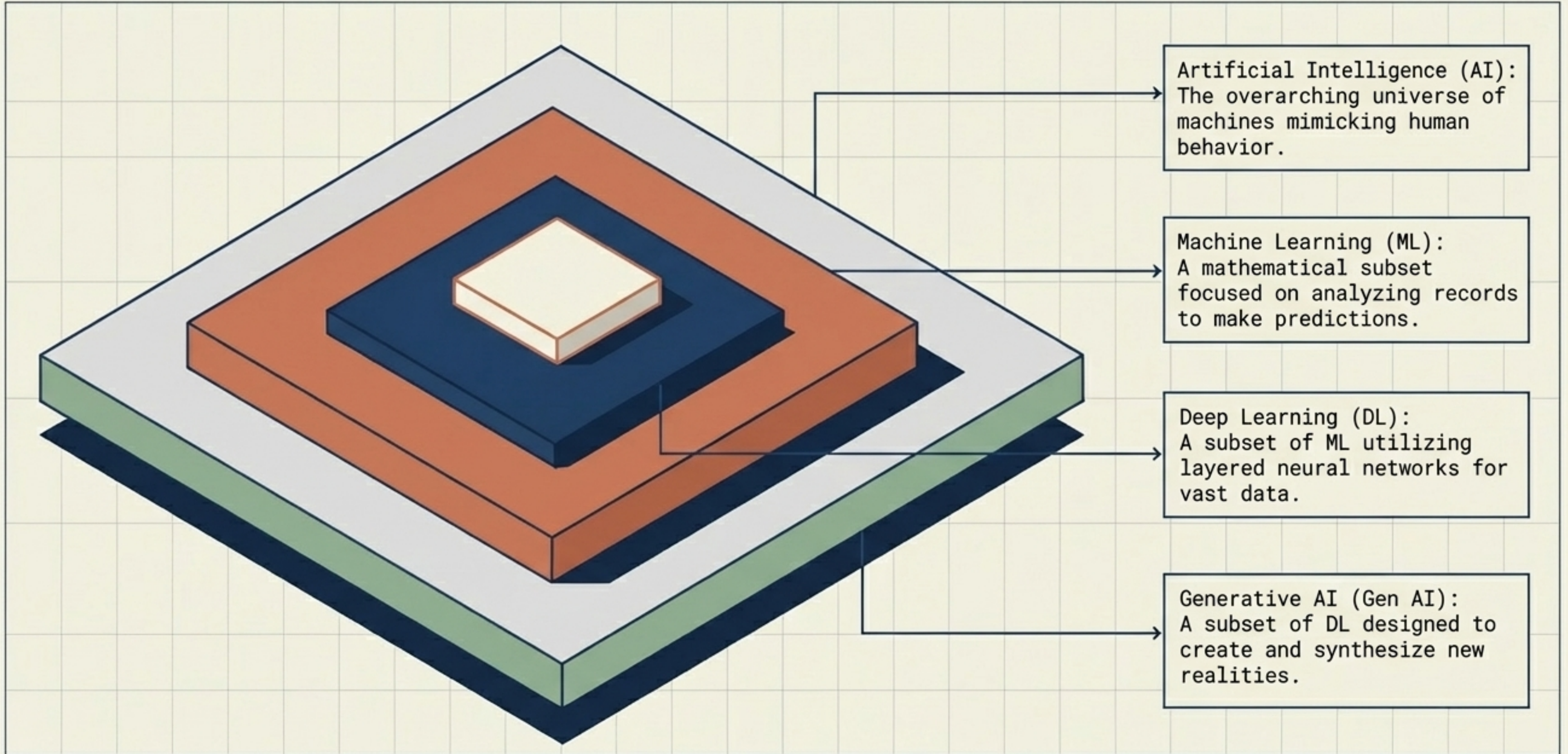


Day 1: Mapping the AI Ecosystem
from Predictive Models to Generative Systems

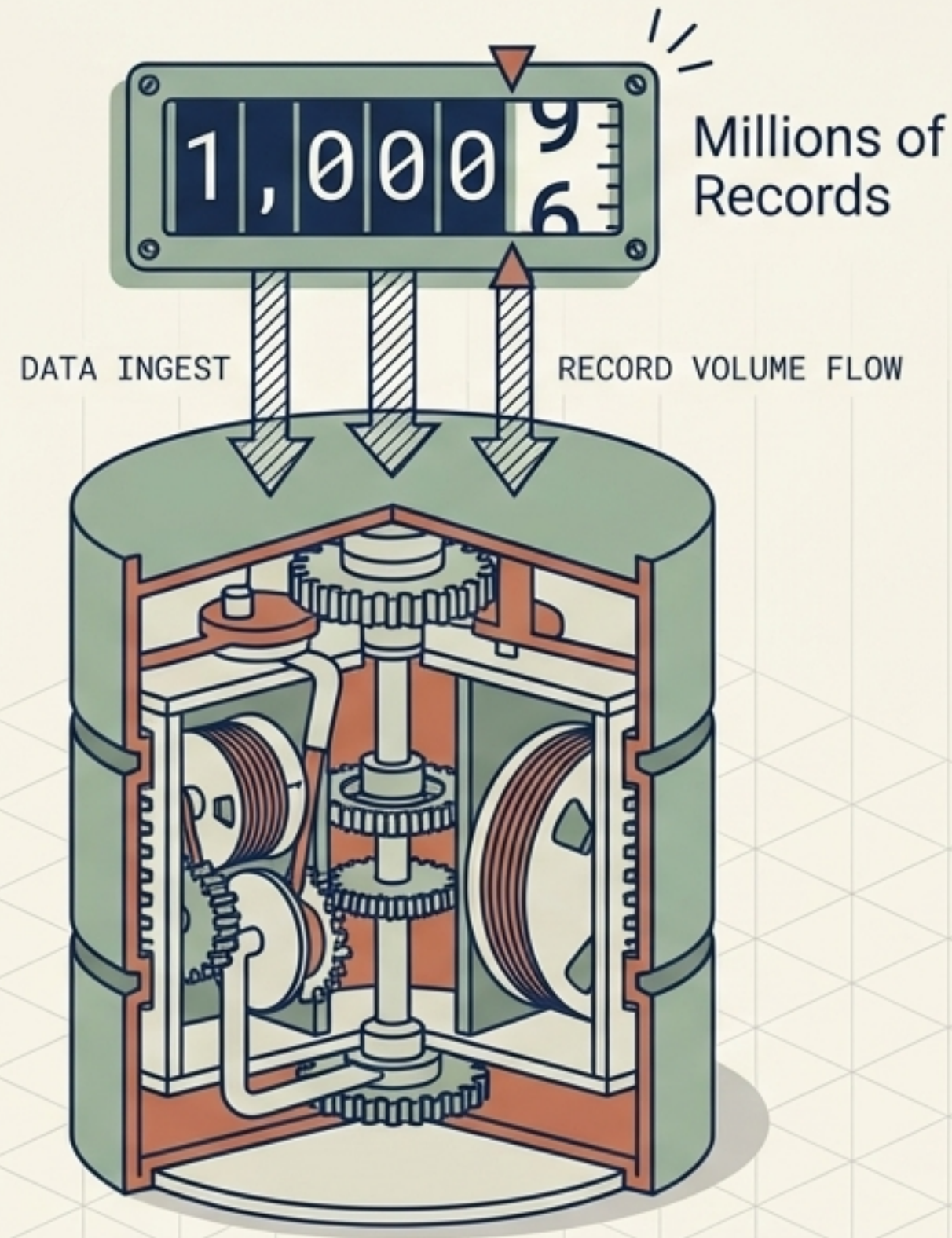
Mimicking Human Behavior: The Predictive Engine



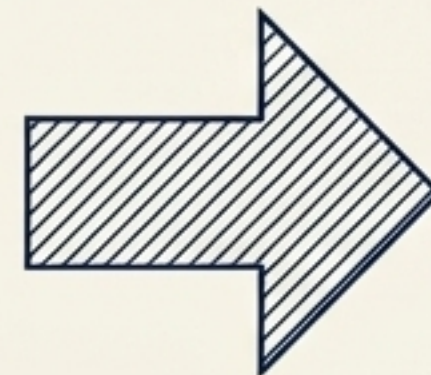
The Nested Taxonomy of Intelligence



Machine Learning (ML): The Art of Prediction



A subset of AI that analyzes 1,000s to millions of records to extract patterns and calculate merriweather probabilities.



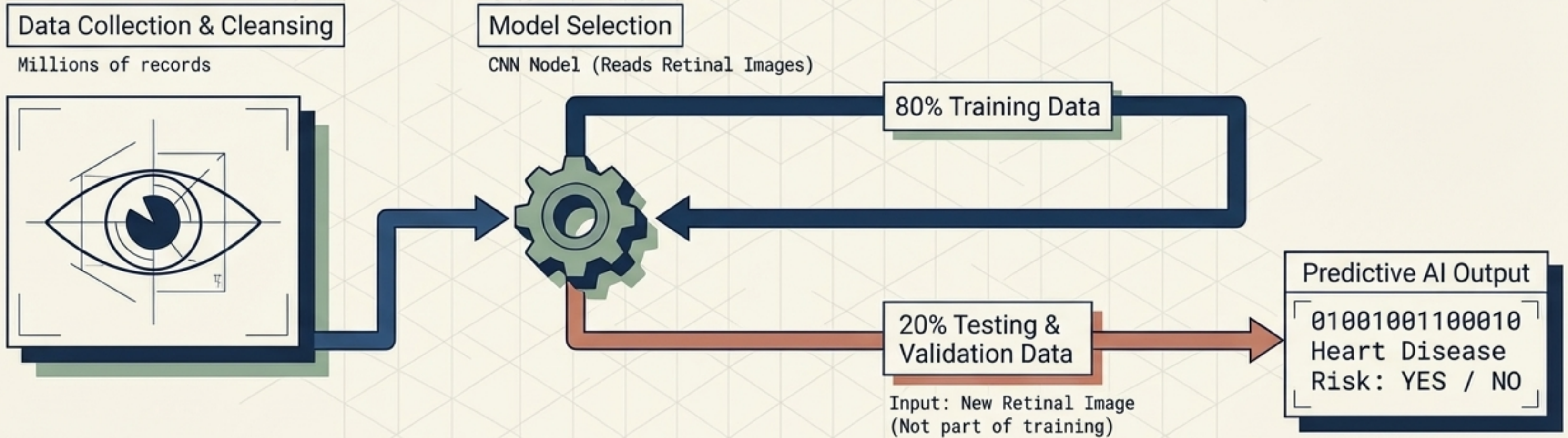
Core Output:
PREDICTIVE AI

Translating Reality into Data: Dataset Anatomy

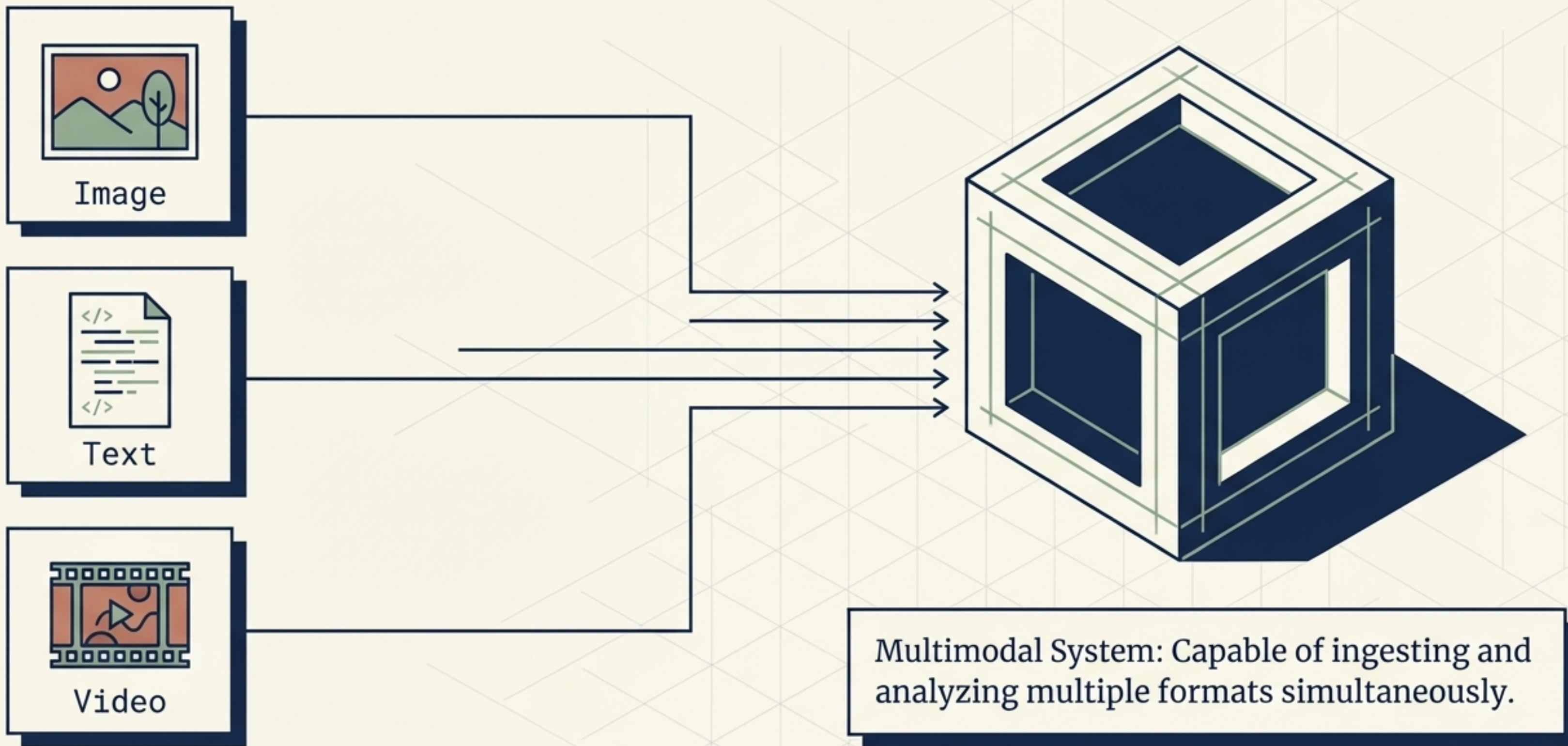
Prediction of Heart Disease

		Parameters / Columns							
Records / Millions of Rows									
	Age	Blood Group	Diabetic	Blood Pressure	Location	Phone	Genetic History	Smoker	Drinker
	✓	✓	✓	✓	✓	✓	✓	✓	✓
	✓	✓	✓	✓	✓	✓	✓	✓	✓

The Machine Learning Pipeline: From Image to Insight



The Multimodal Matrix



Machine Learning Typologies: How Systems Learn

Supervised Learning	Unsupervised Learning
Labelled Data Mechanism: Uses explicit labels (e.g., Content-based Star Ratings). User Interaction: Learns from explicit Likes and Dislikes.	Unlabelled Data Mechanism: Analyzes hidden behaviors and genres. User Interaction: Collaborative Filtering (infers preferences with NO explicit likes or comments).

Supervised Learning Subtypes: Numbers vs. Categories

Supervised Learning

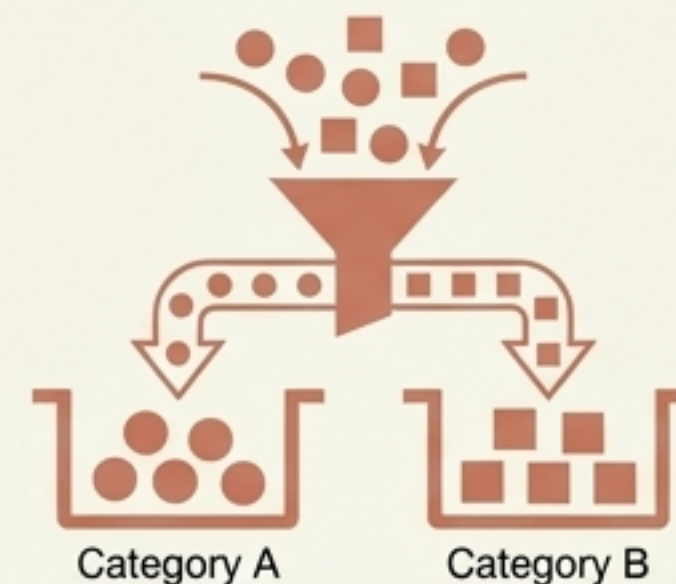
Linear Regression



Output Type: Numerical Value.

Real-World Use Cases: Real Estate (Home purchase pricing), Defect Prediction Testing.

Classification

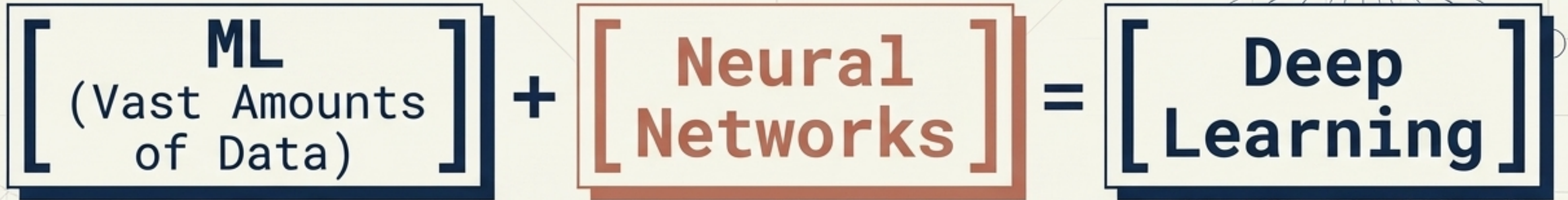


Output Type: Labels or Categories.

Real-World Use Cases: Google predicting Heart Attack Risk (Category: Yes or No), Defect Classification Prediction.

Deep Learning (DL): Scaling the Network

Helvetica Neue



Retains the core
goal:
Highly advanced
Predictive AI.

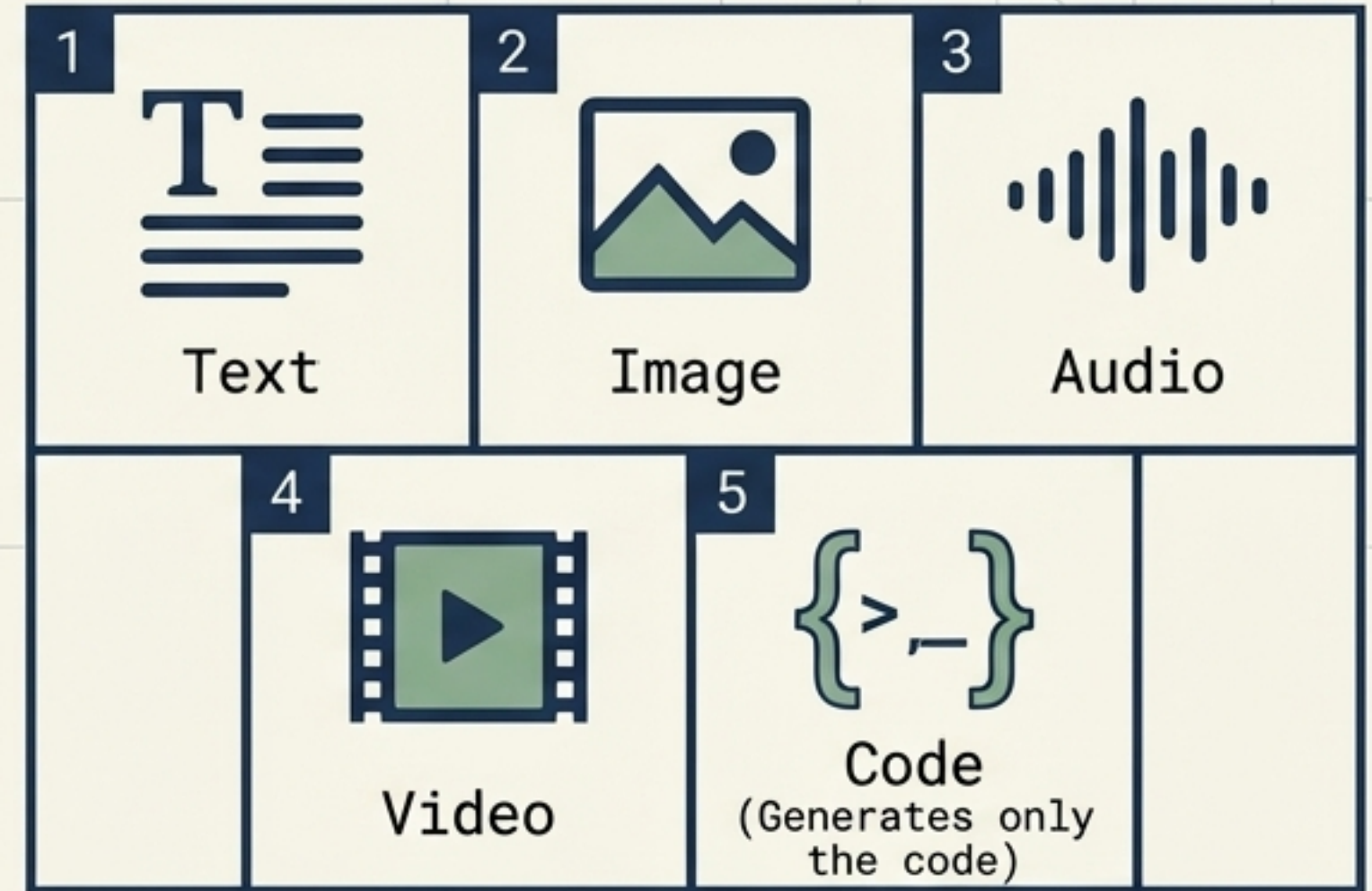
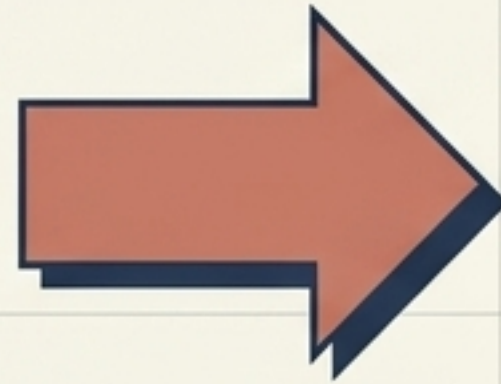
Generative AI: The Paradigm Shift

A subset of Deep Learning.

The fundamental shift: Gen AI is NOT predictive; it is Generative.

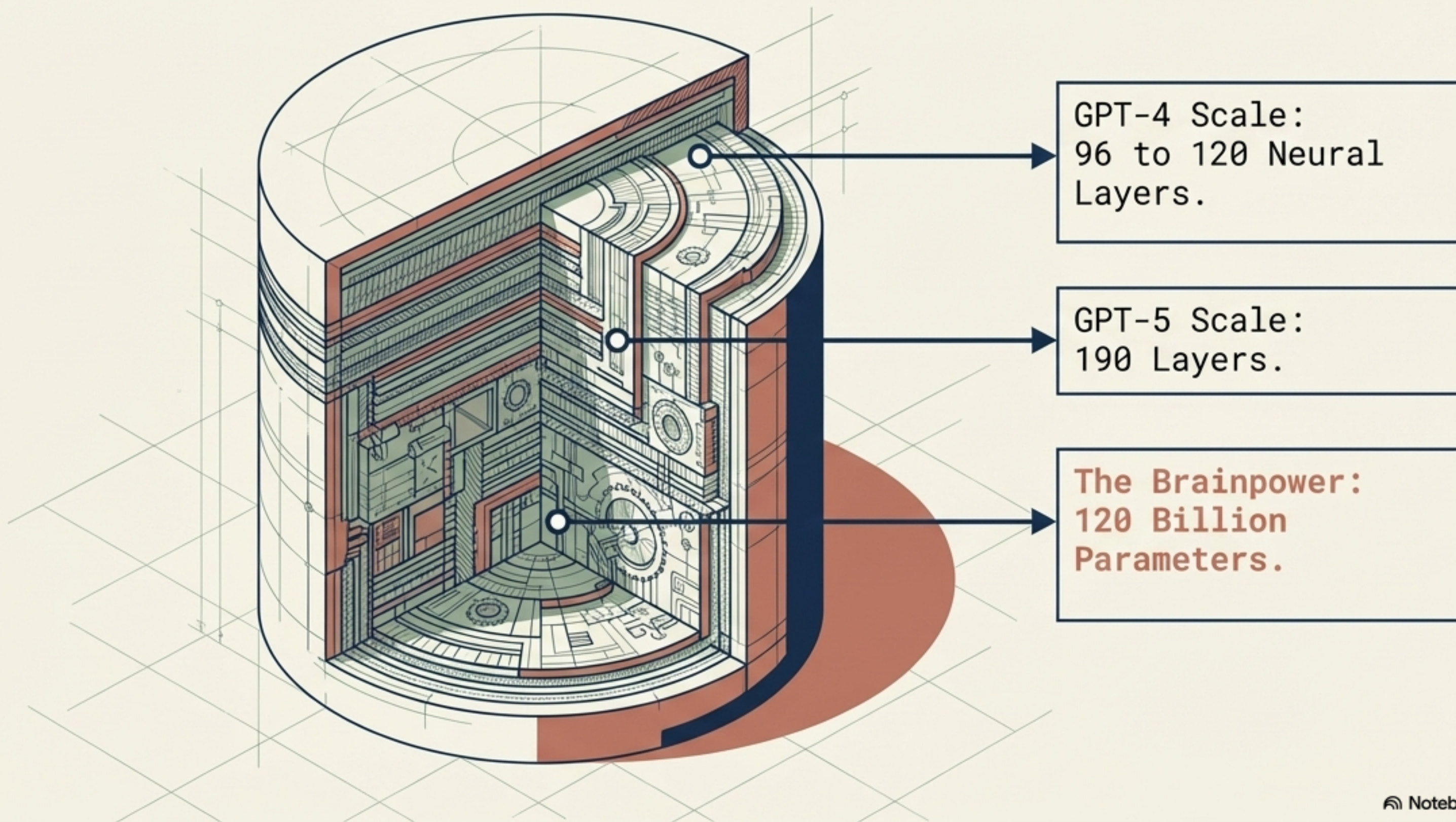


Predictive AI



Anatomy of a Generative Pretrained Transformer (LLM)

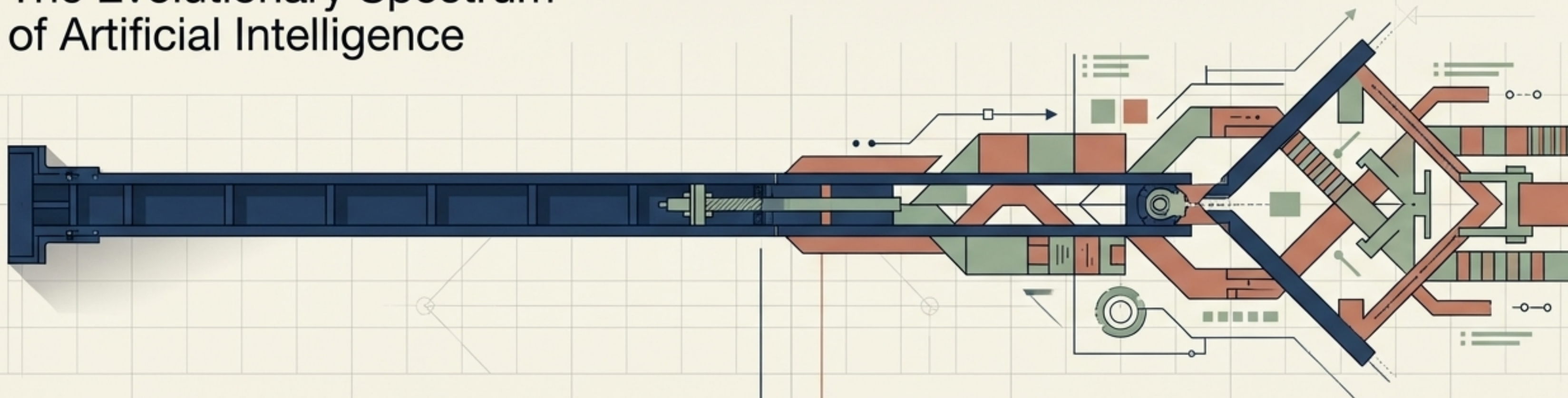
Pretrained with huge datasets across ALL fields, not just software.



The Paradigm Matrix: Predictive vs. Generative Systems

	Machine Learning / Deep Learning	Generative AI
The Core Action	Predictive AI (Analyzes data to forecast outcomes).	Generative AI (Synthesizes data to create new content).
Data Scale	1,000 to Millions of records.	Trillions of records (Pre-trained across all fields).
Engine Type	Mathematical Algorithms + Neural Nets.	Pre-trained Transformers (LLMs).

The Evolutionary Spectrum of Artificial Intelligence



The Analyzers

Components: Machine Learning & Deep Learning.

Components: Machine Learning & Deep Learning.

Drive: Historical Data & Extracted Features.

Ultimate Goal: To Understand and Predict Reality (e.g., What is the fastest route? or Is there a heart risk?).

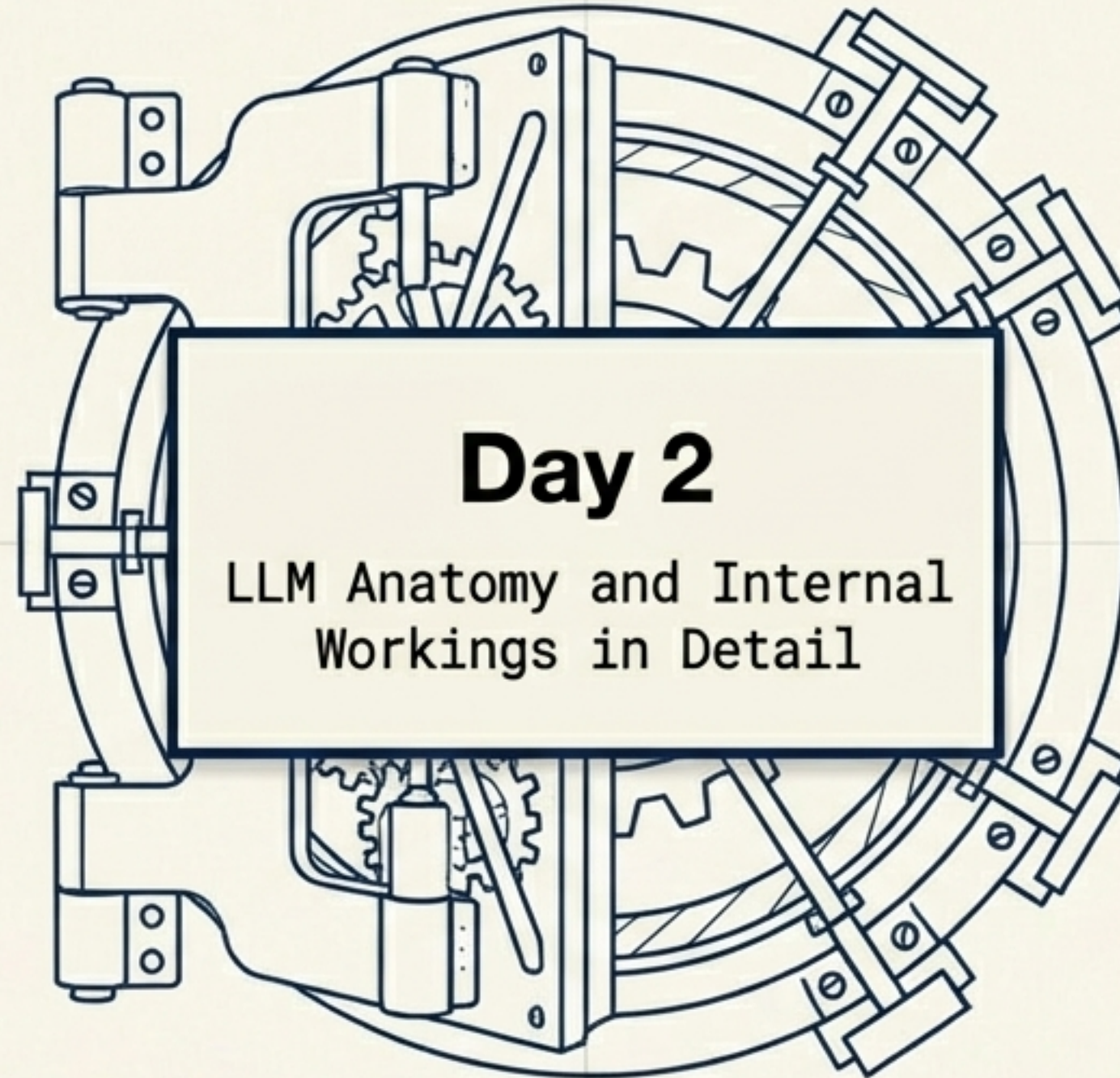
The Creators

Components: Generative AI & LLMs.

Drive: Vast Neural Networks & Billions of Parameters.

Ultimate Goal: To Synthesize and Generate New Realities (e.g., Writing code, generating images).

Next in the Architecture...



End of Day 1 Foundations. Systems initialized.